

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 32 | Date: 12-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (6 Questions - 12 Marks)

Q1. [2 marks] (Easy)

Which of the following data structures stores elements in a contiguous memory location?

Options:

A) Linked List B) Tree C) Array D) Graph

Q4. [2 marks] (Easy)

The operation to add an element to a stack is commonly known as:

Options:

A) Enqueue B) Push C) Insert D) Add

Q5. [2 marks] (Medium)

What is the postfix expression for the infix expression $A * (B + C) - D$?

Options:

A) $ABC+D-$ B) $ABC+D-$ C) $ABC+D-$ D) $ABC+D-$

Q7. [2 marks] (Easy)

In a Queue data structure, elements are removed from which end?

Options:

A) Front B) Rear C) Middle D) Any end

Q9. [2 marks] (Hard)

Which of the following is NOT a characteristic of a Binary Search Tree (BST)?

Options:

A) The left subtree of a node contains only nodes with keys lesser than the node's key. B) The right subtree of a node contains only nodes with keys greater than the node's key. C) Both the left and right subtrees must also be binary search trees. D) The tree must always be perfectly balanced for optimal performance.

Q10. [2 marks] (Medium)

What is the time complexity of searching for an element in a sorted array using Binary Search?

Options:

A) $O(n)$ B) $O(\log n)$ C) $O(n \log n)$ D) $O(1)$

Section 2: Short Answer (4 Questions - 20 Marks)

Q2. [5 marks] (Medium)

Explain the primary difference between a Data Structure and an Abstract Data Type (ADT) with an example for each.

Q3. [5 marks] (Hard)

Describe the worst-case time complexity of the Bubble Sort algorithm. Provide a scenario where Bubble Sort performs in its worst-case.

Q6. [5 marks] (Hard)

Write a short C-like pseudocode function to insert a new node at the beginning of a singly linked list. Assume the list is represented by a pointer to its head node.

Q8. [5 marks] (Medium)

Briefly explain two advantages of using a Linked List over a contiguous Array for storing a collection of data.

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